

ICWaR Course

WR 204

Title: Aqueous Geochemistry (2:1)

Instructor: Prof. Praveen C. Ramamurthy

Co-Instructor: Dr. Soumita Boral

Term: January - May

Course description: The focus of this course is to develop an understanding of the chemical compositions of natural waters, emphasizing both physiochemical and biogeochemical processes operating in aqueous systems: equilibrium vs. kinetic controls, redox reactions, aqueous complexes, chemical weathering, solute sources, and mass balances in watersheds, etc.

Moreover, the application of stable and radiogenic isotope geochemistry used for fingerprinting processes that control water chemistry (rivers, oceans, groundwater) will be discussed. Simple hydrogeochemical modelling techniques will also be introduced.

This three-credit course will be offered as an elective at ICWaR and aimed as a graduate-level course. There are no pre-requisites for this course.

Syllabus:

1. Introduction to the hydrosphere: water cycle, catchment hydrology mass balance, biogeochemical cycles
2. Aqueous chemical reactions: Equilibrium and kinetic concepts, acids & bases, redox, adsorption-desorption, speciation and complexes, organic reactions and interactions.
3. Isotope geochemistry: stable isotopes (O, H, C and N), stable metal isotope systems (Ca, Mg, Li, Fe, etc.), radiogenic isotopes (Rb-Sr, etc.) applications in natural aquatic systems.
4. Weathering: silicate and carbonate chemistry
5. Composition of natural waters: rivers, oceans, groundwater, global issues
6. Hydrogeochemical modelling techniques: mixing models, source apportionments, etc.

Books:

1. Donald Langmuir (1997), Aqueous Environmental Geochemistry. Prentice Hall.
2. Carol Kendall and Jeffrey J. McDonnell (Eds) (2006), Isotope Tracers in Catchment hydrology. Elsevier.
3. William M. White (2013), Geochemistry. Wiley-Blackwell.
4. William M. White (2015), Isotope Geochemistry. Wiley-Blackwell.